

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Added integration tests and monitored production logs across releases and integrations, debugging incidents to root cause and shipping fixes to keep the multi-tenant backend reliable.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Implemented backend security controls, AWS KMS encryption, auth hardening, and rate limiting, remediating **20+ VAPT findings**, and built internal tooling used daily by the AI team.
- Served as primary technical POC and piloted a COO-sponsored GitHub Copilot experiment to automate tech-debt and code-migration workflows, onboarding a second developer.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with input validation and clear contracts used in production by the frontend team.
- Ramped quickly in an existing codebase, migrating repositories from JavaScript to TypeScript, and shipped a production feature using a **4-bit quantized Llama-7B** model via llama.cpp.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with cron-automated workflows and git-webhook auto-deploy, independently end to end; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, pytest, SQLite

- Hardened a Python/FastAPI service with **150+ automated tests (pytest)**, provider routing, and pipeline instrumentation to catch failures across backends, turning open-ended queries into structured, provenance-tracked tables via typed retrieval planning and deterministic extraction with LLM fallback.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Used **Spark UI dashboards** and stage-level metrics to debug shuffle and skew bottlenecks in a pipeline over a **~1.4B-row** dataset, cutting end-to-end runtime **~25%** and the tail-latency ratio from **2.14x to 1.74x** with broadcast joins, skew-join handling, partitioning, and column pruning.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a system integrating camera and microphone **hardware** with a FastAPI backend and React frontend into one offline device pipeline, MediaPipe hand tracking, gesture classification, whisper.cpp ASR, and local TTS, with async APIs and queues keeping it responsive under live streams.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, C++, JavaScript, TypeScript, Java, SQL, Bash

Test & CI: pytest, integration testing, test automation, CI/CD, GitHub Actions, debugging

Systems & Ops: monitoring, logging, instrumentation, automation, Docker, Linux

Data & Web: PySpark, Spark UI, matplotlib, pandas, FastAPI, React, Next.js

Cloud & Platforms: AWS, GCP, Azure, Git, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.